

VANESSA SCLEAROVA

vanessasclearova@gmail.com ◇ Web: <https://vanessik.github.io/> ◇ Scholar: [7ez3M0AAAAAJ](#)

Computer Vision & AI researcher with 6+ years of experience working on diverse research & engineering projects. At the moment, a Ph.D. student at ETH Zurich, ([CVG group](#)) and Max Planck Institute for Intelligent Systems ([Perceiving Systems](#)) with a focus on digital humans.

PROFESSIONAL EXPERIENCE

ETH Zurich, ([CVG group](#))

Research Scientist, Ph.D. student

07/2025 - Present

- Research on dynamic head avatars from multi-view images.

Max Planck Institute for Intelligent Systems, ([Perceiving Systems](#))

Research Scientist, Ph.D. student

06/2023 - Present

- 3D reconstruction, rendering, and simulation of head avatars with strand-based hair.
- Developed a text-guided generative model and single-view strand reconstruction method.
- Research on hair rendering with Gaussian Splatting.
- Published 3 first-author papers (CVPR 2024, ICCV 2025, 3DV 2026) + 7 collaborative works.
- Supervised 5 students and 4 group projects on computer vision and virtual reality.

Samsung AI Center

Research Engineer

08/2021 - 06/2023

- Worked on mesh reconstruction methods for learning human hair geometry;
- Co-authored a paper accepted to ECCV 2022 on one-shot mesh reconstruction of human avatars;
- Co-authored a patent on reconstruction and rendering of human head avatars ([#12169900](#));
- Worked on multi-view hair reconstruction methods with strand-based representations;
- Co-authored a paper accepted to ICCV 2023 on strand-based hair reconstruction for human avatars.

Samsung AI Center

Assistant Engineer

06/2021 - 08/2021

- Developed a Reptile-based meta-learning framework for NeRF models;
- Integrated the meta-learning pipeline into the top-performing model in distributed mode.

Huawei

Assistant Engineer

06/2020 - 10/2020

- Tested state-of-the-art low-bit quantization methods on Transformer and CNN/MLP models;
- Conducted experiments on automatic quantization using reinforcement learning.

Gazprom Neft

Assistant Engineer

02/2019 - 01/2020

- Developed reinforcement learning agents (DQN and DQfD) for solving dynamic pricing problems;
- Implemented multi-armed contextual bandit algorithms;
- Achieved over 90% of the total possible rewards with developed agents;
- Co-authored a patent on dynamic pricing ([#2022619566](#));

EDUCATION

ETH Zurich

Zurich, Switzerland

Ph.D. in Computer Science

09/2023 - Present

Topic: *Digital human avatars*, prof. Marc Pollefeys, prof. Michael Black, prof. Justus Thies

Skolkovo Institute of Science and Technology, [Skoltech](#)

Moscow, Russia

M.Sc. in Computer Science (GPA 4.84 / 5.00)

Topic: *Reinforcement learning for Deep brain simulation* ([thesis](#)), prof. Dmitry Dylov

Moscow Institute of Physics and Technology, [MIPT](#)

Moscow, Russia

M.Sc. in Computer Science (GPA 4.95 / 5.00)

Moscow Institute of Physics and Technology, [MIPT](#)

Moscow, Russia

B.Sc. in Applied Mathematics and Physics (GPA 4.4 / 5.00)

Topic: *Mathematical model of HIV infection*, Institute of Numerical Mathematics, prof. Gennady Bocharov

MOCHI: Registration-free Multi-view Face Prediction	2026
P. Filntisis, G. Retsinas, R. Danecek, <u>V. Sklyarova</u> , P. Maragos, T. Bolkart	CVPR
PhysHead: Simulation-Ready Gaussian Head Avatars (page)	2026
B. Kabadayi, <u>V. Sklyarova</u> , W. Zielonka, J. Thies, G. Pons-Moll	CVPR
GeomHair: Reconstruction of Hair Strands from Colorless 3D Scans (page)	2026
R. Lazuardi, A. Sevastopolsky, E. Zakharov, M. Nießner, <u>V. Sklyarova</u>	CVPR workshop
How to Build Digital Humans? From Priors to Photorealistic Avatars (page)	2026
W. Zielonka*, T. Kirschstein*, T. Bolkart*, S. Giebenhain, <u>V. Sklyarova</u> , X. Deng, et al.	Eurographics (STAR)
Vid2Haircut: 3D Strand-Based Hairstyle Reconstruction from Video (page)	2026
F. Ben Ayed, G. Becherini, J. Thies, <u>V. Sklyarova</u>	Eurographics (Short Papers)
NeuralFur: Animal Fur Reconstruction from Multi-view Images (page)	2026
<u>V. Sklyarova*</u> , B. Kabadayi*, A. Yiannakidis, G. Becherini, M. Black, J. Thies	3DV
<i>(Oral presentation, Best Paper Runner Up Award)</i>	
Im2Haircut: Single-view Strand-based Hair Reconstruction for Human Avatars (page)	2025
<u>V. Sklyarova</u> , E. Zakharov, M. Prinzler, G. Becherini, M. Black, J. Thies	ICCV
Joker: Conditional 3D Head Synthesis with Extreme Facial Expressions (page)	2025
M. Prinzler, E. Zakharov, <u>V. Sklyarova</u> , B. Kabadayi, J. Thies	3DV
Gaussian Haircut: Human Hair Reconstruction with Strand-Aligned 3D Gaussians (page)	2024
E. Zakharov, <u>V. Sklyarova</u> , M. Black, G. Nam, J. Thies, O. Hilliges	ECCV
HAAR: Text-Conditioned Generative Model of 3D Strand-based Human Hairstyles (page)	2024
<u>V. Sklyarova</u> , E. Zakharov, O. Hilliges, M. Black, J. Thies	CVPR
Neural Haircut: Prior-Guided Strand-Based Hair Reconstruction (page)	2023
<u>V. Sklyarova</u> , J. Chelishev, A. Dogaru, I. Medvedev, V. Lempitsky, E. Zakharov	ICCV
<i>(Oral presentation, top 1.8% of the submissions)</i>	
Realistic one-shot mesh-based head avatars (page)	2022
T. Khakhulin, <u>V. Sklyarova</u> , V. Lempitsky, E. Zakharov	ECCV

* denotes equal contribution.

AWARDS

Team hackathon prizier	Erlangen, Germany
<i>Present a solution for automatic paper reproduction using the ChatGPT system.</i>	Dec 2023
Entrepreneurial Spirit certificate (link)	Russia, Moscow
<i>Skoltech Entrepreneurship Award;</i>	June 2022
Finalist of accelerator competition	Russia, Moscow
<i>Our startup idea came to the final in the accelerator competition.</i>	Feb 2022
Team hackathon prizier (3rd place) (link)	Russia, Moscow
<i>The task from Sbermarket: recommend top 50 future products for each user.</i>	Sep 2020
Team hackathon winner (1st place) (link)	Russia, Moscow
<i>The task from Tinkoff: develop a game with finance context using Oleg (voice chatbot).</i>	Sep 2020
Team case winner (1st place)	Russia, Moscow
<i>"StudStock", The task from Vkusvill: build a recommendation system.</i>	Apr 2019
Team hackathon prizier (2nd place)	Russia, Dolgoprudny
<i>The task from Gazpromneft: maximize profit on gas stations, using reinforcement learning.</i>	Oct 2018

SKILLS / AREAS

Digital Humans · 3D Modeling and Reconstruction · 3D Scanning · 2D & 3D Computer Vision · 3DMMs · NeRFs · Gaussian Splatting · Neural Rendering · Nerf · Diffusion Models · Hair and fur reconstruction · Physics-based simulations · Computer Graphics for Vision · Reinforcement Learning · Python · C/C++ · PyTorch · Blender

LANGUAGES

English (full professional), German (basic), Russian (native)